

Alexander Sebastian Christianto

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LinkedIn: www.linkedin.com/in/alexandersc22 | Portfolio: github.com/alexandersc22/portfolio

EDUCATION

Nanyang Technological University, Singapore

Bachelor of Engineering in Electrical & Electronic Engineering

With Specialization in Data Analysis & Machine Learning

And Minor in Business. Honours (Distinction), CGPA:4.04

Jul 2022 – Jun 2026

- Courses: Artificial Intelligence & Data Mining, Machine Learning Design & Application, Semiconductor Fundamentals, Analog Electronics, Business Finance, Fundamentals of Management, Marketing.

Professional Certifications

- Coursera: Google Data Analytics Professional, Google Business Intelligence Specialization, Business intelligence and data analytics: Generate insights (by Macquarie University).

EXPERIENCES

Data Analyst Intern at Systems on Silicon Manufacturing Company (SSMC)

Jan 2025 – May 2025

- Evaluated and verified incoming material quality within the Supplier Quality Management team, ensuring strict adherence to semiconductor manufacturing standards.
- Conducted in-depth analysis of supplier performance metrics and material quality compliance utilizing Excel and Tableau.
- Designed and deployed interactive dashboards for real-time performance monitoring, improving data visibility for multiple cross-functional teams across the company.

NTU Christian Fellowship Honorary General Secretary

May 2024 – May 2025

- Partnered with TOPS management to oversee overall club operations and drive strategic organizational decisions.
- Streamlined club administration by managing internal communications, coordinating schedules, and maintaining comprehensive meeting documentation to ensure seamless operations.

PROJECTS

Business Intelligence & Data Analytics Project: Cyclistic Bike-Share Analysis

Mar 2026 – May 2026

- Engineered an end-to-end business intelligence solution to analyze millions of NYC Citi Bike trip records, aimed at identifying customer demand and informing strategic station expansion.
- Formulated comprehensive Stakeholder, Project Requirements, and Planning documents to define metrics and align cross-functional leadership on business objectives.
- Developed SQL queries and executed ETL processes over a 6-week project lifecycle to integrate primary trip data with secondary Census boundary and weather datasets.
- Designed an accessible, interactive dashboard in Tableau to visualize key performance indicators, including year-over-year trip growth, seasonal usage trends, and station congestion limits.

Final Year Project

AI-Driven Topological Structured Light for Advanced Optical Communication

Aug 2025 – May 2026

- Developed AI algorithms for optimized generation of diverse topological light structures using SLMs.
- Created neural network models for real-time prediction and compensation of atmospheric turbulence effects on OAM beams.
- Developed a real-time simulator for an adaptive sender AI that automatically switches from multiplexed to single-mode when link health drops below a 95% threshold.

Other Data Analytics Projects (more details in my portfolio at github.com/alexandersc22/portfolio)

- Hotel Ratings & Trip Type Analysis: Analyze customer feedback through data visualizations.
- Instagram Clone Database: Perform advanced analysis on a robust relational database model using SQL.
- NBA 2020 Season Analysis: Data cleaning and modification tasks to ensure smooth analysis process.
- Premier League Match Analysis: Investigate PL dataset to uncover insights about the teams' performance.

SKILLS

Languages : Proficient in English, Indonesian, and Malay. Basic Chinese.

Technical Skills : Python Data Analysis, SQL, R, Tableau, Power BI, SAS Viya, Looker Studio, Microsoft Excel.